Lab Session: Dimension Reduction & Latent Variables

From Standard PCA to Probabilistic Models

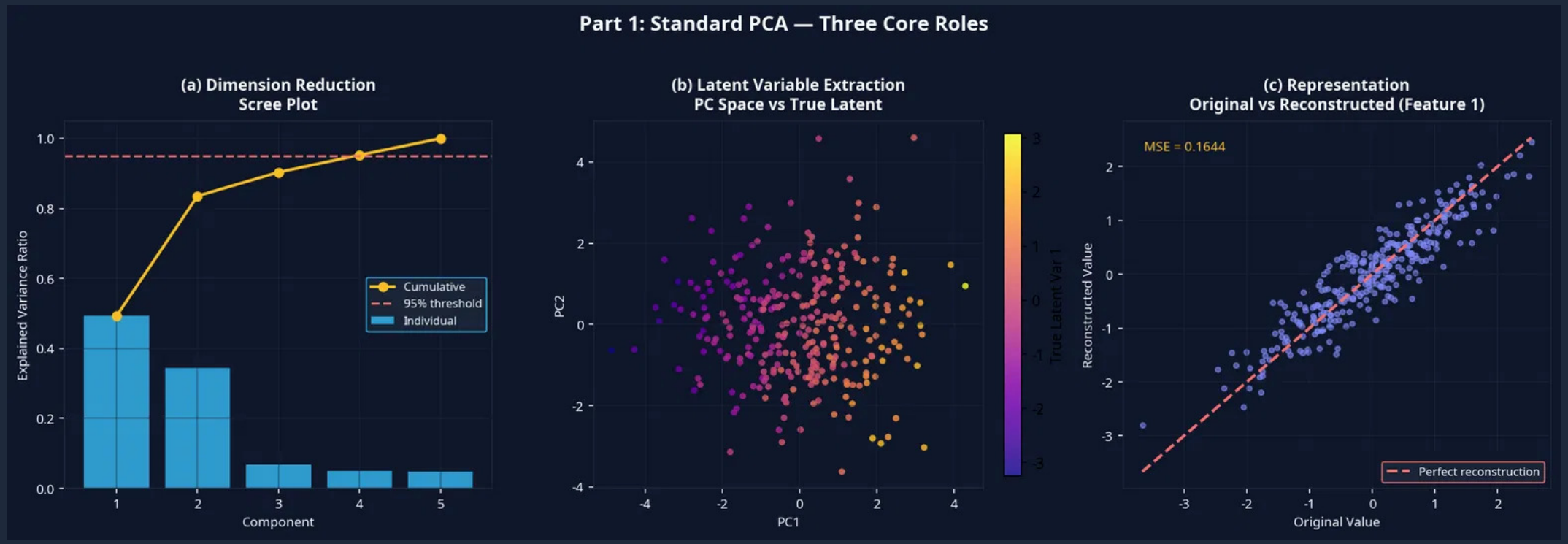
INTERACTIVE PYTHON DEMONSTRATIONS

Lab Overview and Goals

In this interactive Python lab session, we will explore the practical applications and limitations of Principal Component Analysis (PCA) and its advanced variants. The session is divided into four main parts, followed by a student coding exercise.

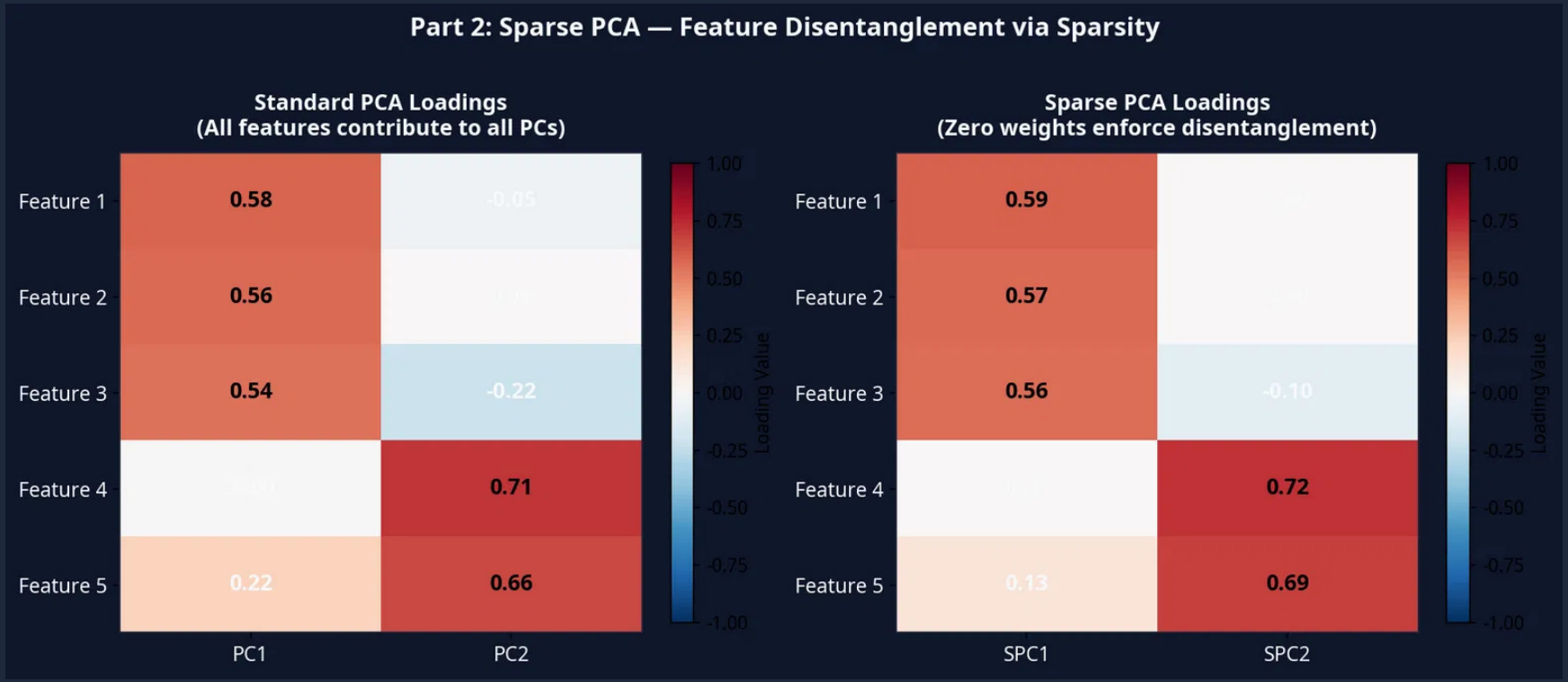
Part 1: Standard PCA

We begin by demonstrating the three primary roles of Principal Component Analysis on a synthetic dataset generated from two known latent variables.



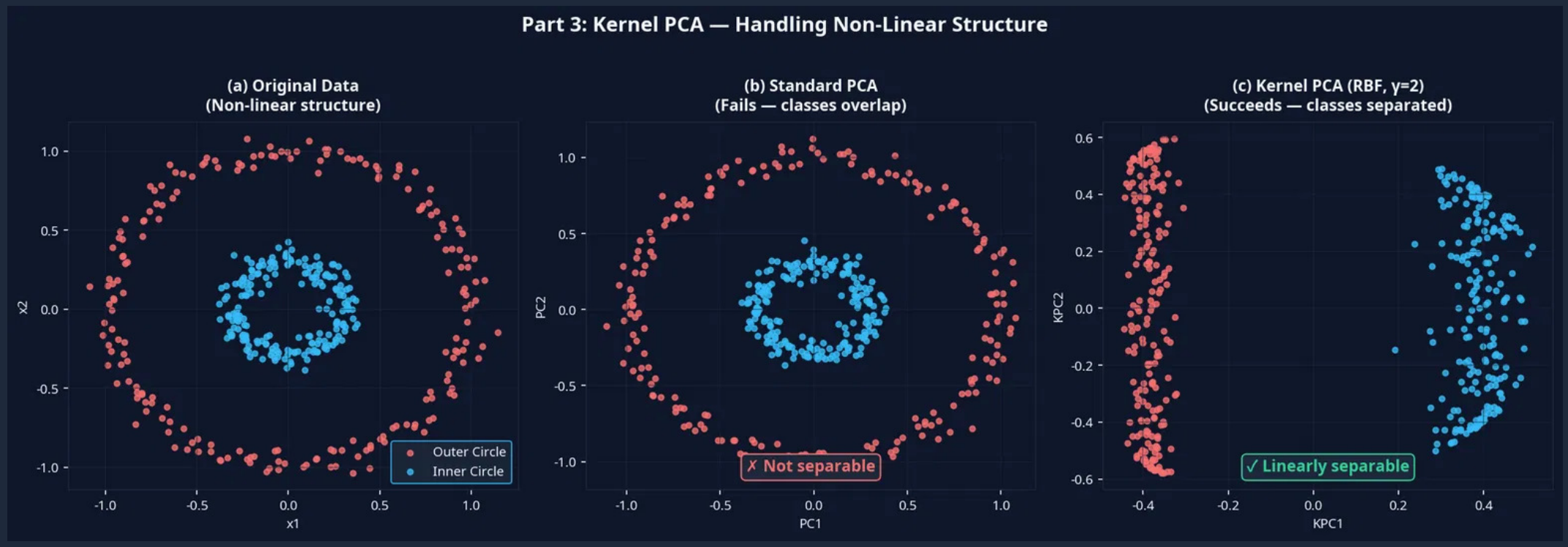
Part 2: Sparse PCA

Standard PCA creates components that are linear combinations of all features, making interpretation difficult. Sparse PCA adds an L1 penalty, forcing many weights to zero and successfully disentangling the underlying features.



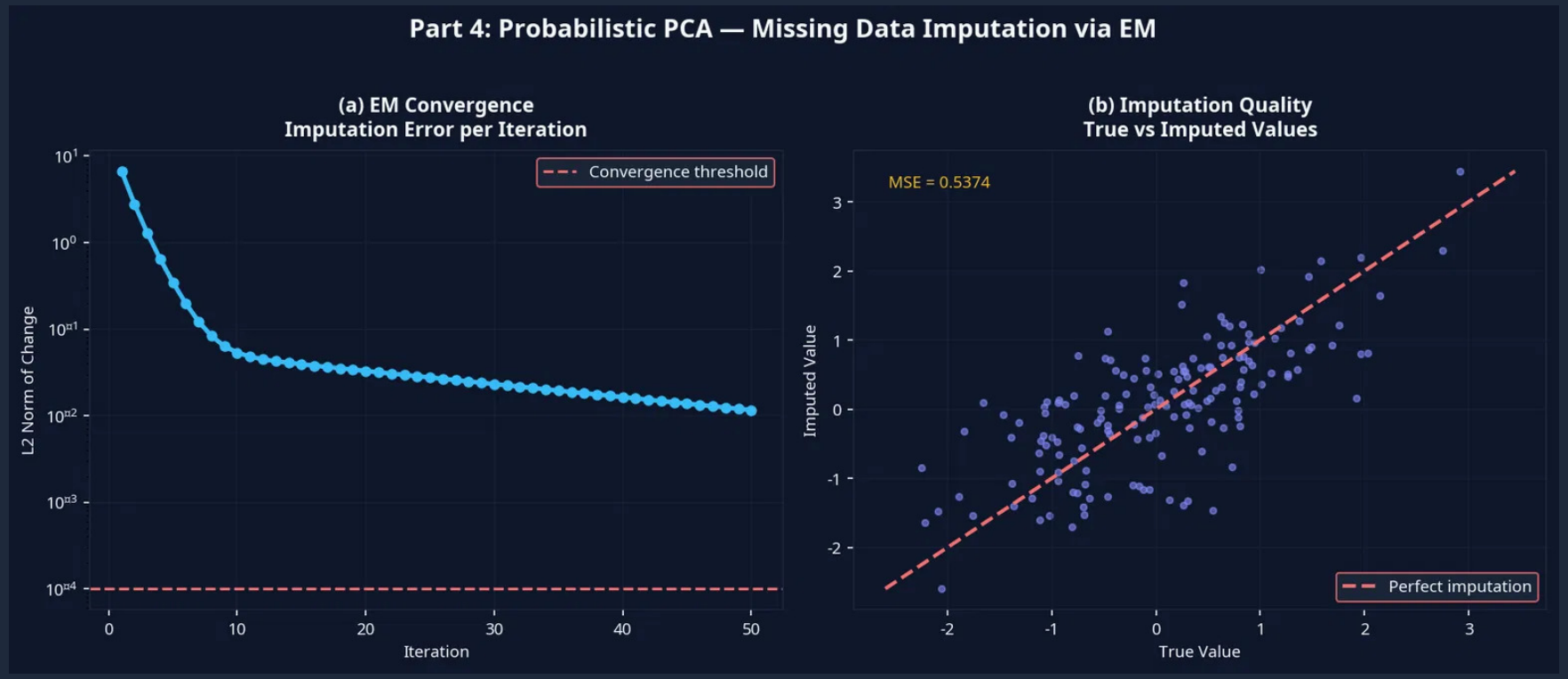
Part 3: Kernel PCA

Standard PCA assumes a linear subspace and fails on non-linear manifolds. By applying the "kernel trick" (e.g., using an RBF kernel), Kernel PCA implicitly maps the data into a high-dimensional space where it becomes linearly separable.



Part 4: Probabilistic PCA

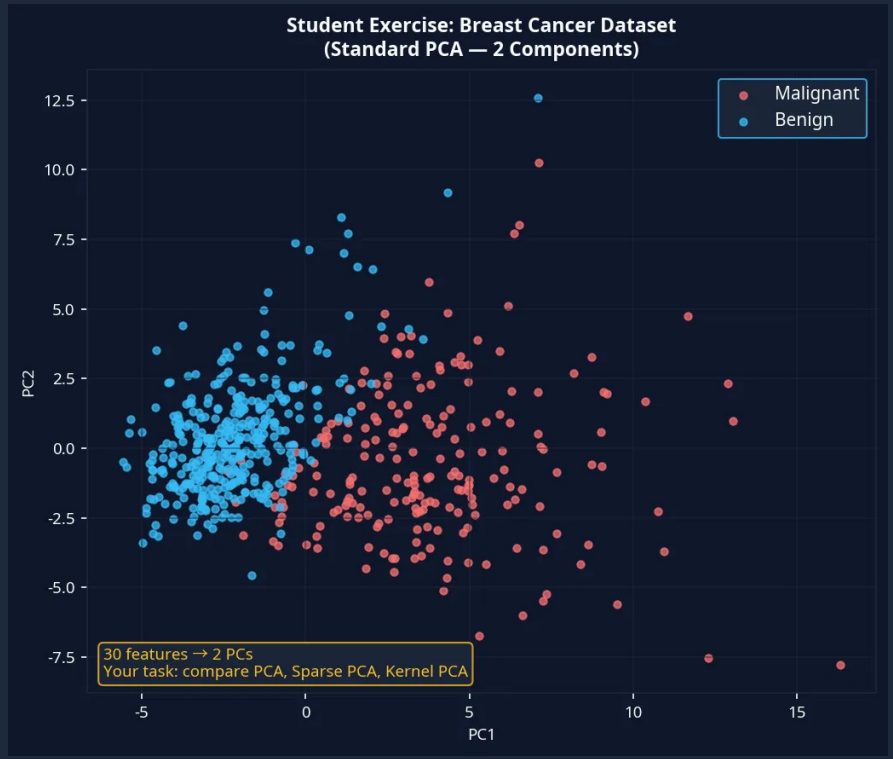
Probabilistic PCA formulates PCA as a generative latent variable model. This probabilistic framework allows us to handle missing data gracefully using the Expectation-Maximization (EM) algorithm.



Part 5: Student Exercise

YOUR TASK

Apply the concepts learned today to the high-dimensional **Breast Cancer Wisconsin dataset** (30 features).

- **Standard PCA:** Reduce to 2 components and plot the result.
- **Sparse PCA:** Examine the components. How many features are exactly zero in the first sparse principal component?
- **Kernel PCA:** Try both `poly` and `rbf` kernels. Does a non-linear kernel provide better visual separation of the classes?

> **Goal:** Compare the three methods and understand their practical trade-offs in a real-world classification setting.